Targets Briefing - Multi-Person UWB Localization

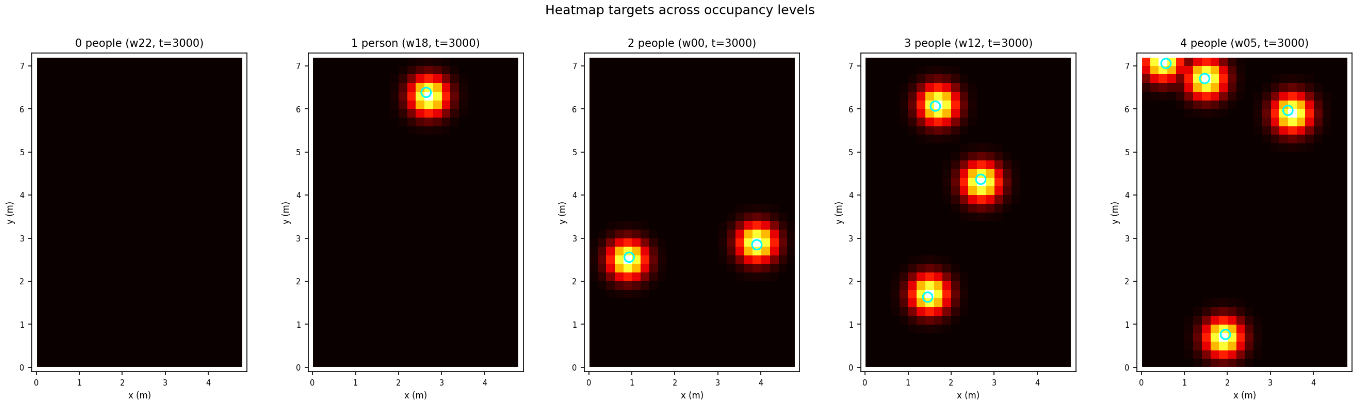
Generated 2026-04-30 12:47 - Hybrid heatmap + count head - Grid 24 x 36 cells of 20 x 20 cm - Sigma 1.5 cells = 30 cm - Temporal context 8 frames (~0.32 s)

1. Target design

Parameter	Value	Rationale
Heatmap shape	(36, 24)	Full room 4.8 x 7.2 m at 20 cm cells
Cell size	20 x 20 cm	Below 1m inter-person spacing seen in 4-people windows
Gaussian sigma	1.5 cells = 30 cm	Wide enough to give the model a smooth gradient signal
Peak placement	Snap to nearest cell, peak = 1.0	Clean BCE target; sub-cell recovered by NMS centroid refinement
Multi-person merge	Element-wise max	Keeps target a presence map (0..1), not a density
Count head classes	5 ({0..4})	Used as predicted k for top-k peak extraction at inference
Temporal context	8 frames (~0.32 s)	p99 motion 0.86 m/s -> ~14 cm in 0.32 s, < 1 cell

2. Heatmap targets across occupancy levels

One representative frame from each occupancy class (0--4 people). Cyan circles are the ground-truth (x, y) markers, the heatmap shows the generated target. Peaks sit at the centre of the cell containing each person; their amplitude is exactly 1.0.



3. Round-trip xy -> heatmap -> peaks -> xy

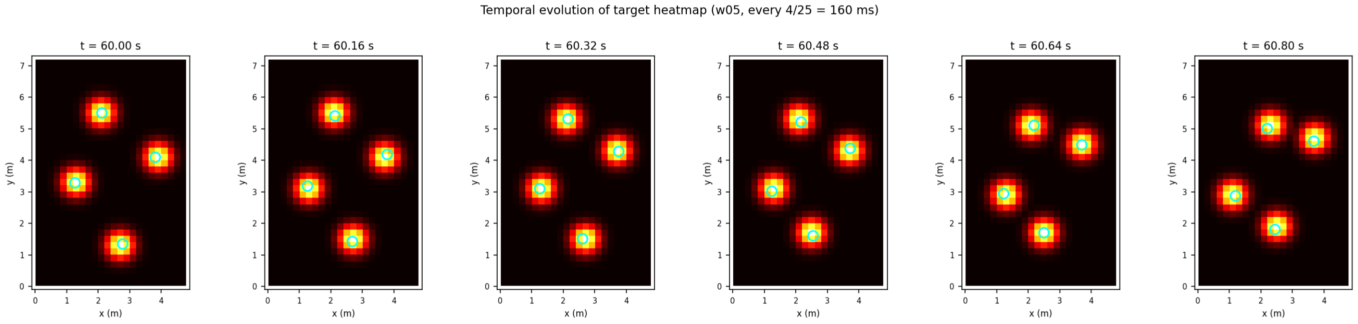
Generated targets for 50 consecutive frames of a 4-person window (w05), then ran the peak extractor with k = ground-truth count and matched extracted peaks to ground-truth positions greedily. The error floor reflects the snap-to-cell quantization (half-cell ~ 10 cm); on real model output the predicted heatmaps will be asymmetric and the 3x3 centroid refinement will recover sub-cell position.

Statistic	Value (cm)
Mean error	7.82
Median error	8.21
P90 error	11.18
Max error	12.95
Samples	200

(missing: targets_round_trip.png)

4. Temporal smoothness

Six consecutive frames, 160 ms apart, showing target heatmaps drifting smoothly across cell boundaries as subjects walk. No abrupt jumps at cell transitions.



5. Frame stacking (model input)

At inference time the model receives the most recent 8 preprocessed frames. The stacking is causal: stack[t] = preprocessed[t-7..t]. The first 7 frames of each window are warmed up by replicating the first frame.

Quantity	Value
Per-frame preprocessed shape	(6, 3, 105, 2)
Stacked input shape	(8, 6, 3, 105, 2)
Stacked input elements	30,240
Stacked input size (float32)	118.12 KB
Heatmap output shape	(36, 24, 1)
Heatmap output size (float32)	3.38 KB
Count output shape	(5,) softmax